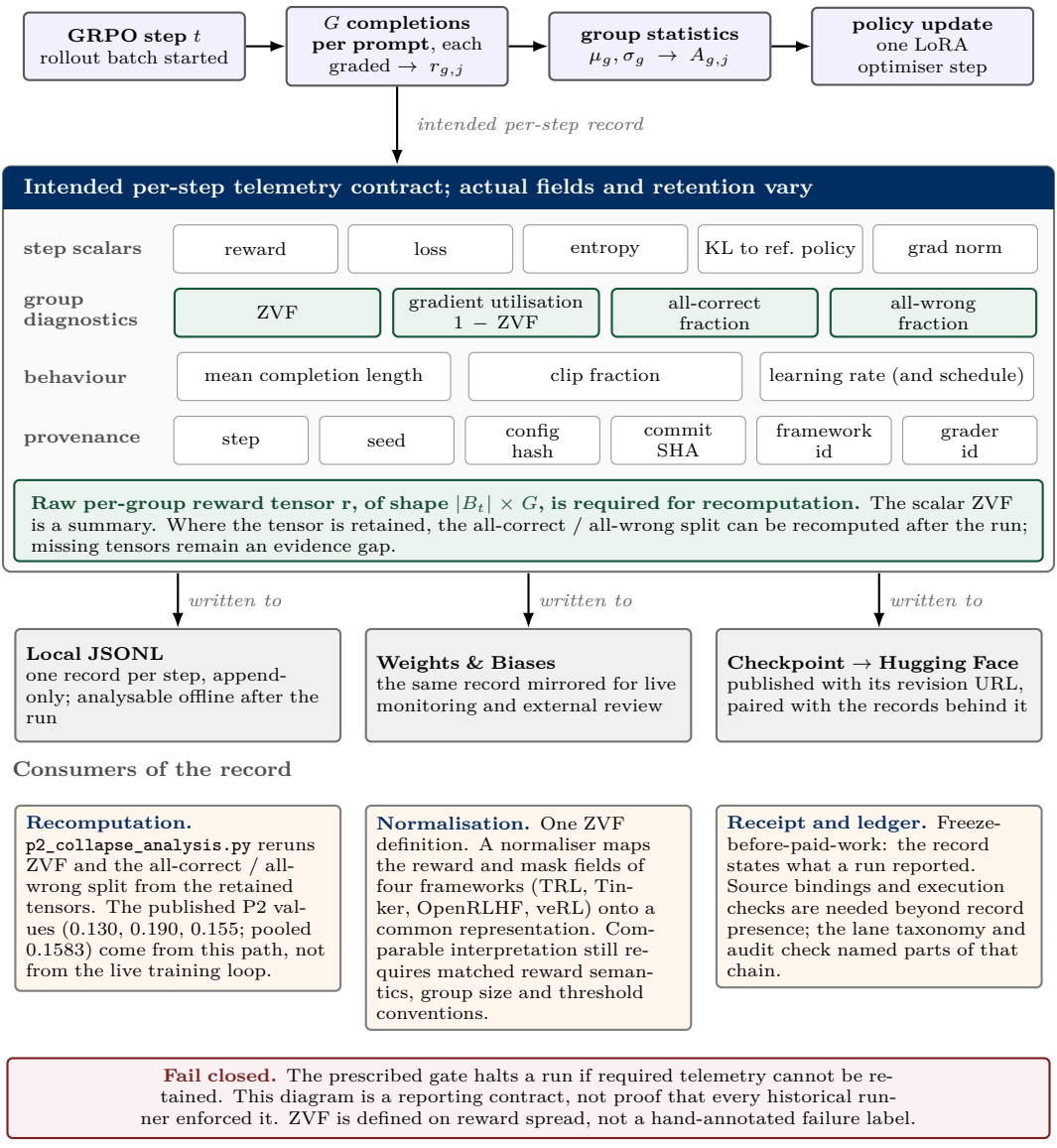
GRPO step t
rollout batch started
G completions per prompt, each graded → $r_{g,j}$
group statistics $\mu_g, \sigma_g \rightarrow A_{g,j}$
policy update
one LoRA optimiser step
intended per-step record
Intended per-step telemetry contract; actual fields and retention vary
step scalars
reward
loss
entropy
KL to ref. policy
grad norm
group diagnostics
ZVF
gradient utilisation 1 − ZVF
all-correct fraction
all-wrong fraction
behaviour
mean completion length
clip fraction
learning rate (and schedule)
provenance
step
seed
config hash
commit SHA
framework id
grader id
Raw per-group reward tensor r, of shape $|B_t| \times G$, is required for recomputation. The scalar ZVF is a summary. Where the tensor is retained, the all-correct / all-wrong split can be recomputed after the run; missing tensors remain an evidence gap.
written to
written to
written to
Local JSONL
one record per step, append-only; analysable offline after the run
Weights & Biases
the same record mirrored for live monitoring and external review
Checkpoint → Hugging Face
published with its revision URL, paired with the records behind it
Consumers of the record
Recomputation.
p2_collapse_analysis.py reruns ZVF and the all-correct / all-wrong split from the retained tensors. The published P2 values (0.130, 0.190, 0.155; pooled 0.1583) come from this path, not from the live training loop.
Normalisation. One ZVF definition. A normaliser maps the reward and mask fields of four frameworks (TRL, Tinker, OpenRLHF, veRL) onto a common representation. Comparable interpretation still requires matched reward semantics, group size and threshold conventions.
Receipt and ledger. Freeze-before-paid-work: the record states what a run reported. Source bindings and execution checks are needed beyond record presence; the lane taxonomy and audit check named parts of that chain.
Fail closed. The prescribed gate halts a run if required telemetry cannot be retained. This diagram is a reporting contract, not proof that every historical runner enforced it. ZVF is defined on reward spread, not a hand-annotated failure label.